2 A grenade is launched at ground level with velocity of 75 m s^{-1} and angle 20^0 from the horizontal to hit a sniper on top of a 30 m tall building at a position 245 m from the foot of the building. The sniper is standing at a distance *x* from the edge of the building.

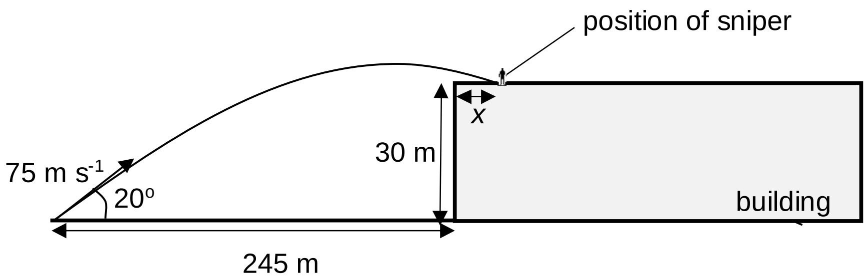

Fig. 2.1

(a) Show that the time for the grenade to reach the sniper is 3.5 s. [1]

$$s_y = u_y t + \frac{1}{2} g t^2$$

$$30 = 75 \sin 20^o t - \frac{1}{2} x 9.81 t^2$$

$$t = 3.5 \ s$$

(b) Calculate the speed of the grenade just before impact. [2]

$$v_y = u_y + at$$

$$= 75 \sin 20^o - \frac{1}{2} x 9.81 x 3.5$$

$$= 8.5 \ ms^{-1}$$

$$v_x = 75 \cos 20^o = 70.5 \ ms^{-1}$$

$$v = \sqrt{8.5^2 + 70.5^2} = 71 \ ms^{-1}$$

(c) Determine the distance, *x*, of the sniper from the edge of the building. [2]

$$u_x t = 245 + x$$

$$75 \cos 20^o x 3.5 = 245 + x$$

$$x = 1.7 \ m$$

(d) The sniper runs away immediately from the edge of the building when the grenade is launched. Given that the grenade has a 'kill radius' of 10 m, calculate the minimum constant acceleration at which the sniper should run in order to escape. [1]

$$s = u t + \frac{1}{2} a t^2$$

$$10 = \frac{1}{2} a.(3.5)^2$$

$$a = 1.6 \ ms^{-2}$$

For Examiner's Use

3 A peg is fixed to the rim of a vertical turntable of radius r, rotating with a constant angular speed